Fiscal Year:	FY 2013	**Task Last Updated:**	FY 07/27/2012
PI Name:	Kelton, Kenneth F. Ph.D.		
Project Title:	NASA Research under ESA-Based Investigations THERMOLAB and ICOPROSOL		
Division Name:	Physical Sciences		
Program/Discipline:	MATERIALS SCIENCE		
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	63130	**Congressional District:**	1
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2009 ESA AO Biology on Sounding Rockets and Physical Sciences on All Platforms
Start Date:	10/01/2010	**End Date:**	09/30/2015
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	3	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA MSFC
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Flight Program:	ISS		
Flight Assignment:	NOTE: To be extended to 9/30/2014 per PI (Ed., 7/10/13)		
Key Personnel Changes/Previous PI:	Kenneth Kelton is U.S. Team Member on ESA-sponsored projects, “Thermophysical properties and solidification behaviour of undercooled Ti-Zr-Ni liquids showing an icosahedral short-range order (ICOPROSOL),” and "Thermophysical Properties of Liquid Metallic Alloys - Modelling of Industrial Solidification Processes and Development of Advanced Products (THERMOLAB - ISS)."		
COI Name (Institution):			
Grant/Contract No.:	NNX10AU19G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Funding is for Dr. Kelton's role as Team Member on the subprojects described below, associated with ESA-sponsored main projects as noted. Main project: Thermophysical and structural properties and solidification behaviour of supercooled Vit106 and Vit106a glass forming liquids as a member of THERMOLAB; the Science Coordinator is Prof. Dr. Hans Fecht, Ulm University, Institute of Micro and Nanomaterials, Albert-Einstein-Allee 47, Ulm Germany. Similar measurements on undercooled Ti-Zr-Ni liquids showing an icosahedral short-range order (ICOPROSOL); the Science Coordinator is Dr. Dirk Holland-Moritz, Deutsches Zentrum für Luft- und Raumfahrt (DLR), Institut für Materialphysik im Weltraum, Linder Höhe, Köln, Germany. Subprojects: 1. Ground-Based Experimental Programme. Synchrotron X-ray studies of the liquid structure will be made as a function of the Ni concentration at the Advanced Photon Source using the ESL chamber at Washington University, which has been optimized for scattering studies. These will be correlated with the thermophysical property data obtained on the MSL-EML. Terrestrial maximum undercooling and induction time to recalescence data will be obtained in ESL, where stirring ensures that nucleation is not diffusion controlled. This data will allow an evaluation of diffusion in the MSL-EML undercooling studies. 2. Thermophysical Property Measurements - ISS. Specific heat, density, surface tension and viscosity measurements will be made of supercooled Ti-Zr-Ni liquids with different Ni concentrations as a function of the temperature using the MSL-EML facility. These will be correlated with x-ray synchrotron diffraction studies of the supercooled liquids. The purpose is to determine the existence and nature of liquid/liquid phase transitions that may be associated with chemical and topological ordering. 3. Effect of diffusion on nucleation. Nucleation studies will be performed using the MSL-EML facility. The critical items of interest are the maximum undercooling and the induction time to recalescence in isothermal holds at temperatures below the liquidus temperature. Together with the results from the thermophysical property measurements and the MHD convection model results this will allow us to analyse how the maximum undercooling is influenced by fluid flow. These results of the ìg experiments will be compared with the results of ground-based ESL measurements. Main project: Thermophysical Properties of Liquid Metallic Alloys - Modelling of Industrial Solidification Processes and Development of Advanced Products (THERMOLAB - ISS); Science Coordinator is Professor Hans-Joerg Fecht, Ulm University, Ulm, Germany Subprojects: Ground Based Experiment Programme. Measurements of the density, surface tension and viscosity and synchrotron x-ray diffraction measurments of the liquid structure in the supercooled state and ground-based ESL measurments of the maximum undercooling and induction time to recalescence in isothermal holds. Long Duration Microgravity Experiments. Measurements of thermophysical properties, specific heat (modulated measurements), surface tension, viscosity and density, and also the electrical resistivity will made as a function of temperature. These will be coupled with x-ray diffraction measurements of the liquid structure factor (WP 2) to study the role of ordering on novel liquid/liquid phase transitions and to determine the nature of these transitions. Maximum undercooling and time to recalescece measurements as a function of temperature and stirring will be compared with terrestrial ESL measurements to assess the importance of diffusion on nucleation, allowing Kelton's coupled flux model for nucleation to be evaluated experimentally.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Product advancement in industries such as information technology, aerospace, automotive, and basic and new materials manufacturing requires advanced materials development. In the past, this was often carried out through a laborious process of guided trial and error. The vast increases in computing power have now opened the door for realistic modeling approaches, which has the potential to greatly improve the efficiency of materials development. However, this is frequently hampered by the lack of relevant thermophysical property data needed for the modeling. Further, key processes such as crystal nucleation from the liquid or other phase transitions within the liquid, are often poorly understood, making it difficult to realistically model them. In many cases, the experimental data and a deeper understanding of some processes cannot be obtained from terrestrial experiments due to interactions with sample containers at the temperatures of interest. Even containerless based methods, such as electrostatic levitation, cannot be used to accurately investigate these. The ISS experiments planned under the support of this grant will allow these data to be obtained.
	NNX10AU19G is a continuation of an earlier grant (NNX07AK27G). NNX07AK27G was focused primarily on ground-based studies in anticipation of a flight opportunity for a planned ISS experiment (QUASI); the focus of that grant was to support ground-based measurements of structures and thermophysical properties of supercooled high-temperature liquids that would support that flight experiment. The focus of NNX10AU19G is to continue to support the ground-based structural, density and viscosity studies of supercooled liquids, and to support an ISS-flight-based study of specific heat, emissivity, thermal conductivity, viscosity, and nucleation studies in a micro-gravity environment. The THERMOLAB team has opened up one slot each in Batches 1 and 2 for QUASI samples, and QUASI has formally stated interest in several other samples that will be studied by colleagues on other teams. In addition, QUASI was a partner on two recently successful ESA proposals (THERMOLAB and ICOPROSOL), which were chosen for flight opportunities in Batches 3 and 4, and possibly in subsequent batches. As a member of the THERMOLAB team, QUASI will carry out measurements on Vit106a, a Zr-based liquid that easily forms a metallic glass. Recently, problems developed with the cups that were to be used in Batch 1, requiring a redefinition of the samples that would be processed in that batch. As part of this process, QUASI was named principal investigator on a new sample, Vit106, a closely related Zr-based BMG forming alloy to that of Vit106a. This past year, these two grants have provided partial support for the PI, a Research Scientist, three graduate students, and two undergraduate students. Key ground-based activities since the beginning of this grant are listed below. These investigations are aimed at obtaining supporting information that will be used to help design the ISS experiments and to help interpret the results. Activities are also underway to develop the facilities needed to prepare the samples for the ISS

Task Progress:	studies. A. Activities directly linked with upcoming ISS experiment Many activities were carried out over the past year, which are directly focused on supporting the ISS experiments. These include: 1. Support of two Internal Working Group meetings: (i) December 12-14, 2011 at ESTEC and (ii) July 14-16, 2012 at Deutsches Zentrum fur Lüft-und Raumfahrt (DLR) – These meetings are required for ISS experimental planning and exchange of project information within the THERMOLAB team and with other teams. 2. Preparation of samples needed for parameter determination by the Microgravity User Support Center (MUSC) – The ISS-EML experiments require that the coupling parameters for each sample be known. We have prepared and delivered samples for these measurements. The sample preparation methods used are identical to those that will be used to prepare the flight samples. 3. Ground-based studies of the thermophysical properties of Vit106 in the electrostatic levitation facility located at Washington University (WU-BESL) – Since it was only decided to fly Vit106 in late spring 2012, it has been necessary to carry out an accelerated ground-based program. This was primarily focused on specific heat (Cp/e) and specific volume (V) measurements to correlate with a reported liquid-liquid phase transition in this alloy liquid. We have found evidence of maxima in both Cp/e and V, which correlate with the hysteresis in density claimed by others as evidence for a liquid-liquid phase transition. 4. Support of a visit with MUSC on June 11-15, 2012 to prepare the science protocol for ISS measurements for Vit106a and Vit106 – The ISS experiment will consist of 102 cycles in helium and vacuum. In addition to one calibration and three cycles of crystallization at the end, the following experimental studies will be made: a. Measure the specific heat, emissivity and thermal conductivity of the equilibrium and supercooled liquid using modulation calorimetry (17 cycles) b. Measure the surface tension and viscosity of the equilibrium and supercooled liquid using the oscillating drop method (8 cycles) c. Determine the time to crystallization as a function of stirring and measure the solidification velocities (18 cycles) d. Determine the maximum undercooling as a function of stirring and measure the solidification velocities (10 cycles) e. Determine the maximum undercooling as a function of cooling rate and measure the solidification velocities (45 cycles) 5. Measurement of the evaporation rates of Vit106a and Vit106 – These were made as a function of temperature for computation of the coil coating and dust production during processing in the MSL-EML on the ISS. Evaporation data were also obtained for Ti-Zr-Ni liquids that are of interest for future ISS Batches. 6. Design and modification of the arc-melting facility for preparation of the ISS samples – The flight samples will be prepared from elemental materials that are chosen to have the lowest oxygen contamination possible. The preparation of low-oxygen samples also requires that the vacuum quality of the arc-melting facility be acceptable. We have made major modifications to our existing facility to ensure this. These include installing all metal and larger diameter piping to improve pumping speed and reduce outgassing, and replacing a rotary vacuum pump with a turbo-pump/oil-free fore pump combination. B. Ground-based activities to obtain supporting information for ISS experiments Some of the ground-based activities were directed at providing additional data that will help design the ISS experimental studies and help interpret the results. These include the following: 1. Identification of an inverse correlation between thermal expansivity and cohesive energy – Our density measurements, made in the electrostatic levitation facility at Washington University (WU-BESL), of supercooled liquids of Cu-Zr have demonstrated that the volume expansivity is correlated with liquid fragility and can be an indicator for good glass formation. This provides additional insight into how the structures of liquids are related to their thermal properties. Under partial support by this grant, we have also measured the volume expansivities and densities for twenty-five transition metal glass-forming liquids and have discovered an inverse correlation between the expansivity and the cohesive energy. The values of expansivity predicted from this relationship also agree reasonably well with the published data for thirty other transition metal and alloy liquids. We have demonstrated that this correlation can be can be understood from a universal relationship that exists between the liquid surface tension and the cohesive energy. 2. Identification of a fractal character of the liquid and the development of that with supercooling – For liquids and glasses it can be argued that the location of the first peak in the structure factor, S(q1), is proportional to the negative cube root of the specific volume, V (i.e. V to the -1/3 power). Recently, a paper in Nature Materials reported that for glasses the exponent is -0.433 instead of -1/3 This was argued to indicate that the glass has a fractal network with fractal dimension of Df = 1/0.433 = 2.31, which is similar to the measured fractal dimension for quasicrystals. Since a hierarchy of atomic clusters characterizes quasicrystals, these results suggest that metallic glasses may also be characterized by a similar arrangement atomic. Using WU-BESL, we have examined this correlation for metallic liquids that form glasses. The data indicate that liquids near the melting temperature have a higher fractal index, consistent with less clustering. However, liquids at lower temperatures, near the glass transition temperature, have a fractal dimensionality of 2.5, closer to the values measured in the glasses. This indicates the development of a cluster-based structure within the liquid, consistent with the ordering inferred from analysis of our Reverse Monte Carlo fits to the scattering data.
Bibliography Type:	Description: (Last Updated: 05/20/2025)
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